

DUSTBADGE · DESIGN DECISION RECORD

DustBadge TRL 2 review decisions

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	DBG-DDR-001	Version	0.1
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/dustbadge		
Source	docs/decisions/0001-trl2-review-decisions.md @ 5045449		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record the TRL 2 review recommendations adopted for TRL 3 work under Amish's 2026-09-25 instruction, and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** proposed. The recommendations for items D1 to D7 are adopted for TRL 3 work pending Amish's review; items O1 and O2 remain proposed, awaiting Amish.

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-25, /populate) listed seven items as "Proposed, awaiting Amish", each with options and a recommendation. On 2026-09-25 Amish asked for this batch of repos to be taken through the usual process with the instruction "you know the drill, nothing gets past TRL 3". He has not reviewed this repo's items one by one. Every item that carries a recommendation is therefore adopted as recommended for TRL 3 under that instruction and stays open for his review. Items without a recommendation stay "Proposed, awaiting Amish".

■ Options considered

The options for each item are those listed in docs/REVIEW.md (session 2026-09-25, /populate) and in DBG-PRC-001 v0.2, Key design choices.

■ Decision

Table 1. Items adopted for TRL 3.

#	Item	Adopted recommendation	Status
D1	Particle sensor	Option (a): Sensirion SPS30 class with a PM4 output, about \$48. The TRL 2 recommendation also said to characterize it above 1 mg/m ³ on CalRig; CalRig's own TRL 2 design uses incense smoke decaying from 300 to 5 µg/m ³ , so it cannot do this as scoped (see Consequences).	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D2	How silica is shown	Option (a): respirable dust always; RCS only as a labeled estimate once a site silica fraction is entered.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D3	Default limits and alerts	US OSHA and MSHA values by default (25 µg/m ³ action level, 50 µg/m ³ limit, 8 h TWA), configurable for other jurisdictions such as the EU 100 µg/m ³ .	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D4	Sampling mode	Continuous sampling all shift, not duty-cycled.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D5	Data ownership	Logs kept by the worker and shared only by the worker's consent, with a stated rule that they are not used for discipline.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D6	First sector	Surface quarries and stone fabrication first; underground coal and explosive atmospheres excluded. The partner is not chosen (O1).	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D7	Case details	Downward screened inlet, high-visibility yellow front shell, spring clip with strap loop. Used in the TRL 3 model.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review

Table 2. Items that remain open.

#	Item	Status
O1	First co-design partner (worker organization, NGO or university hygiene group). The TRL 2 note left the choice to Amish and made no recommendation.	Proposed, awaiting Amish
O2	How a site silica fraction is obtained where no laboratory is near. DBG-PRB-001 v0.2 proposed one filter sample per site and task analyzed by a partner laboratory, but the item was not in the review note's list, so it is not adopted here. DBG-CAL-001 section F adds that at low dust levels the sample needs a higher flow or two shifts.	Proposed, awaiting Amish

New items raised by the TRL 3 calculations (cell size for R7, the R3 reference sample and target, sun exposure for R11, and the R2 range) are listed in docs/REVIEW.md as "Proposed, awaiting Amish" and are not decided here.

■ Consequences

- `project.yaml` : only the TRL fields change. No budget, pitch or problem change was recommended, so `budget_usd` stays at \$90 and the pitch and problem lines are unchanged.
- DBG-PRB-001, DBG-PRC-001 and DBG-REQ-001 are revised to v0.3. The key design choices in the precis now read as adopted for TRL 3, pending Amish's review. R6's default limits now follow D3; R8 is restated as the inlet within 300 mm of the nose and mouth.
- D1 and CalRig: CalRig (CLR-PRC-001 v0.2) is a smoke chamber for 5 to 300 $\mu\text{g}/\text{m}^3$ and says its results will not transfer directly to mineral dust. DustBadge therefore uses CalRig only for badge-to-badge and drift checks at low levels. Characterizing the SPS30 above 1 mg/m^3 needs a dust generator or a partner laboratory, which is test work beyond TRL 3. This is noted in `docs/REVIEW.md` ; CalRig is not edited.
- The TRL 3 calculations (DBG-CAL-001) led to two changes within these decisions: an SHT4x-class humidity and temperature sensor on the carrier board, so the R11 flag can work, and a charger with a cell thermistor input, so charging is blocked outside 0 to 45 °C. They add \$3; parts now cost \$88.00.
- TRL 4 is on hold by Amish's instruction. Nothing in this record authorizes building or testing.